Multivariable Calculus

MATH 1920 Lecture Notes

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Contents

1 Vectors in Two and Three Dimensions

Question: What information specifies a line?

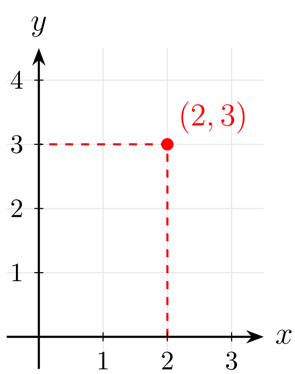
- 2 points
- A point and a direction vector
- Slope and y-intercept
- Intersection of 2 non-parallel planes

Comparison:

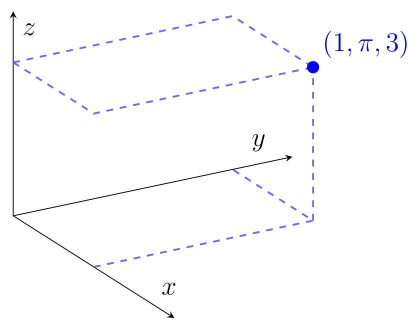
- **Single-variable calculus:** $f(x) \rightarrow$ rate of change of x
- **Multivariable calculus:** $f(x, y) \rightarrow$ functions of multiple variables

1.1 Coordinates

2-dimension (2D): $(2, 3)$



3-dimension (3D): $(1, \pi, 3)$



1.2 Vectors and Position Vectors

Given points $P = (1, 3)$, $Q = (5, 4)$ and $P' = (1, 1)$, $Q' = (5, 2)$:

> **Definition 1.1: Equivalent Vectors**
>
> Two vectors are **equivalent** if one is a translation (pure shift without rotation) of the other.

Example 1.1. Let $R = (4, 1)$ and $O = (0, 0)$. Then:

$$\overrightarrow{PQ} = \overrightarrow{P'Q'} = \overrightarrow{OR}$$

Definition 1.2: Position Vector

A vector whose initial point is at the origin $O(0, 0)$ is called a **position vector**. It is written using angle brackets:

$$\overrightarrow{OR} = \langle 4, 1 \rangle$$

Definition 1.3: Magnitude / Norm

The **length** or **magnitude** of a vector $\vec{v}$, denoted by $\|\vec{v}\|$, is the length of the line segment between its endpoints.

If $\vec{v} = \langle v_1, v_2 \rangle$, then its magnitude is given by:

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2}$$

For example, using $\overrightarrow{PQ} = \langle 4, 1 \rangle$:

$$\left\|\overrightarrow{PQ}\right\| = \sqrt{4^2 + 1^2} = \sqrt{17}$$

1.3 Unit Vectors

A vector $\vec{v}$ is called a **unit vector** if its magnitude is 1, i.e., $\|\vec{v}\| = 1$.

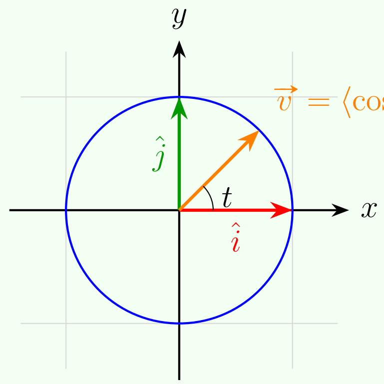
Problem 1.1: Describing Unit Vectors

Describe all unit vectors in $\mathbb{R}^2$.

Solution 1.1

A vector $\vec{v} = \langle x, y\rangle$ is a unit vector if and only if $x^2 + y^2 = 1$. Equivalently, using polar parameterization:

$$\vec{v}(t) = \langle \cos t, \sin t\rangle \quad \text{for } 0 \leq t < 2\pi$$



Standard Basis Vectors:

- **2D Basis Vectors:** $\hat{i} = \langle 1, 0\rangle, \quad \hat{j} = \langle 0, 1\rangle$
- **3D Basis Vectors:** $\hat{i} = \langle 1, 0, 0\rangle, \quad \hat{j} = \langle 0, 1, 0\rangle, \quad \hat{k} = \langle 0, 0, 1\rangle$

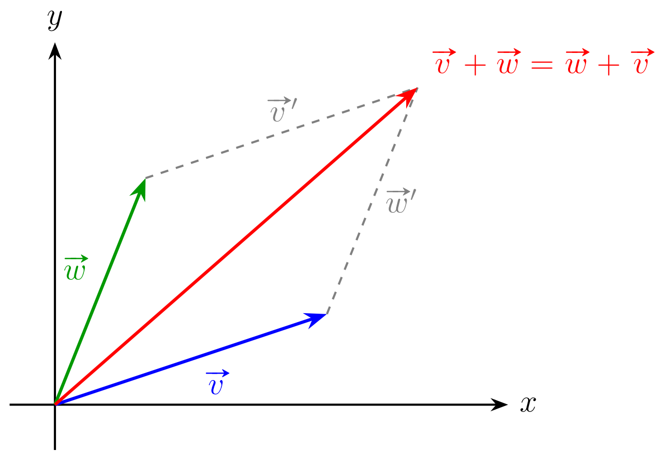
1.4 Vector Operations

Addition and Subtraction:

$$\langle v_1, v_2\rangle + \langle w_1, w_2\rangle = \langle v_1 + w_1, v_2 + w_2\rangle$$
$$\langle v_1, v_2\rangle - \langle w_1, w_2\rangle = \langle v_1 - w_1, v_2 - w_2\rangle \quad \text{(Equivalent to } \vec{v} + (-1)\vec{w}\text{)}$$

Scalar Multiplication: For any scalar $\lambda \in \mathbb{R}$ and vector $\vec{v} = \langle v_1, v_2\rangle$:

$$\lambda\vec{v} = \langle \lambda v_1, \lambda v_2\rangle$$



Important Note

Vectors are **not** designed to be divided! (Vector division is undefined).

1.5 Parallel Vectors

Definition 1.4: Parallel Vectors

Two non-zero vectors $\vec{v}$ and $\vec{w}$ ($\vec{v}, \vec{w} \neq \langle 0, 0 \rangle$) are *parallel* (denoted as $\vec{v} \parallel \vec{w}$) if there exists a non-zero scalar $\lambda \in \mathbb{R}$ such that:

$$\vec{w} = \lambda \vec{v}$$

Equivalent Geometric Definition: $\vec{v} \parallel \vec{w}$ if and only if both vectors define or lie on the *same line passing through the origin.*

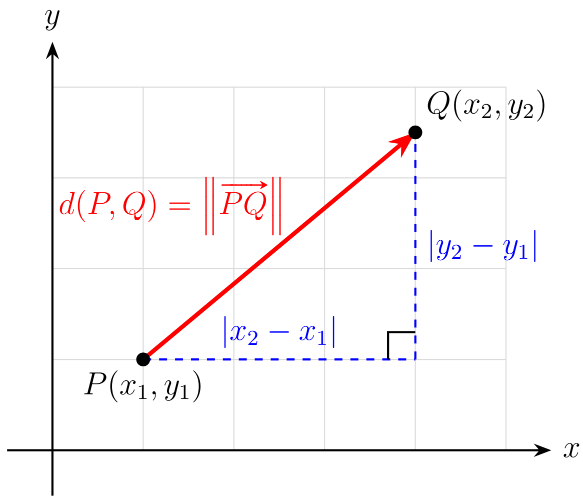
2 Discussion Week 1: Distance Formulae and Geometric Shapes

Definition 2.1: Distance Formula in $\mathbb{R}^2$

Let $(x, y) \in \mathbb{R}^2$ and points $P = (x_1, y_1)$, $Q = (x_2, y_2)$. The distance between P and Q is given by:

$$d(P, Q) = \left\| \overrightarrow{PQ} \right\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This distance formula is a direct application of the Pythagorean theorem, where the distance $d(P, Q)$ forms the hypotenuse of a right-angled triangle with legs $|x_2 - x_1|$ and $|y_2 - y_1|$, as shown in the figure below.



The case in $\mathbb{R}^3$ follows similarly:

Definition 2.2: Distance Formula in $\mathbb{R}^3$

Let $(x, y, z) \in \mathbb{R}^3$ and points $P = (x_1, y_1, z_1)$, $Q = (x_2, y_2, z_2)$. The distance between P and Q is given by:

$$d(P, Q) = \left\|\overrightarrow{PQ}\right\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

2.1 Equations of Geometric Shapes

Consider the equation $x^2 + y^2 = 4$.

In $\mathbb{R}^2$, this equation is equivalent to the set:

$$\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 4\}$$

This represents a circle with radius $r = 2$ centered at the origin. The distance from the origin $(0, 0)$ to any point (x, y) on this circle is constant and equal to 2, which can be verified directly using the distance formula:

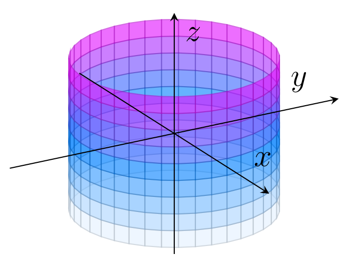
$$\sqrt{(x-0)^2 + (y-0)^2} = 2$$

Hence, the geometry of the circle is directly implied by the Pythagorean theorem.

Now consider the equation $x^2 + y^2 = 1$ in three-dimensional space, $\mathbb{R}^3$. This set is formally expressed as:

$$\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1\}$$

Because z is completely unrestricted, the cross-section at any constant height $z = c$ forms the unit circle $x^2 + y^2 = 1$. Consequently, this equation describes a *right circular cylinder* extending infinitely along the z-axis, as shown.



2.2 Linear Equations across Dimensions

In $\mathbb{R}^2$, the general form of a linear equation is given by:

$$Ax + By + C = 0 \quad (A, B, C \in \mathbb{R}, \text{ with } A \text{ and } B \text{ not both zero})$$

This equation represents a *line* in two-dimensional space.

In $\mathbb{R}^3$, however, the exact same algebraic equation $Ax + By + C = 0$ represents a *plane* parallel to the z-axis (since z is a free variable), not a line.

To uniquely define a line in three-dimensional space, a single linear equation is insufficient; we require a system of *two* independent linear equations (representing the intersection of two non-parallel planes).

3 Vector and Directions

Problem 3.1: Vector Directions

Let $\vec{v} = \langle 3, -4, 2 \rangle$. Find a unit vector in the same direction as $\vec{v}$.

Solution 3.1

To find a unit vector in the same direction as $\vec{v}$, we first compute its magnitude:

$$\|\vec{v}\| = \sqrt{3^2 + (-4)^2 + 2^2} = \sqrt{9 + 16 + 4} = \sqrt{29}$$

The unit vector in the same direction as $\vec{v}$ is then:

$$\hat{v} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{1}{\sqrt{29}} \langle 3, -4, 2 \rangle$$

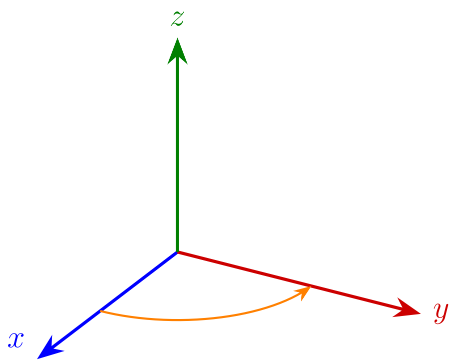
In general, for any non-zero vector $\vec{w}$,

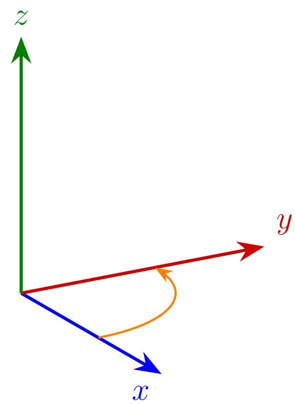
$$\begin{aligned} \|\lambda \vec{w}\| &= \sqrt{(\lambda w_1)^2 + (\lambda w_2)^2 + (\lambda w_3)^2} \\ &= \sqrt{\lambda^2 (w_1^2 + w_2^2 + w_3^2)} \\ &= |\lambda| \, \|\vec{w}\| \end{aligned}$$

Hence, if we want a unit vector in the same direction as $\vec{w}$, we can choose $\lambda = \dfrac{1}{\|\vec{w}\|}$, giving us:

$$\hat{w} = \frac{\vec{w}}{\|\vec{w}\|}$$

Right-Hand Rule The relative position of the x-, y-, and z-axes in $\mathbb{R}^3$ is determined by the *right-hand rule*. If you point your right thumb along the positive x-axis and your index finger along the positive y-axis, then your middle finger (perpendicular to both) will point along the positive z-axis.



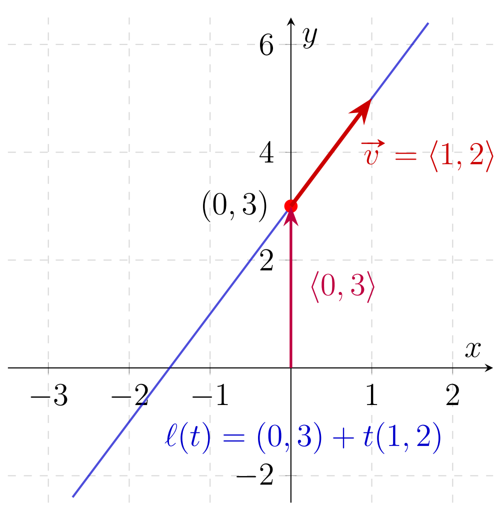


> **Definition 3.1: Parameterizations**
>
> A parameterization is a function whose range is a given geometric object.
>
> For example, a parameterization of the unit circle is given by:
>
> $$f(t) = (\cos t, \sin t)$$

Example 3.1. To parameterize the line $y = 2x + 3$:

$$\begin{aligned} \ell(t) &= (t, 2t + 3) \\ &= \langle 0, 3 \rangle + \langle t, 2t \rangle \end{aligned}$$

The plot is then given:



To generalize, to parameterize a line in $\mathbb{R}^2$, we can first take a point on the line. Then, we take some $\vec{v}$ in the direction of the line. Then, we can write the parameterization as:

$$\ell(t) = (p_1 + v_1 t,\ p_2 + v_2 t)$$

In $\mathbb{R}^3$, we can do the same thing.

Example 3.2. Parameterize the line passing through $P(1, 2, 0)$ and $Q(5, -1, 1)$.

$$\begin{aligned}
\overrightarrow{PQ} &= \langle 5-1, -1-2, 1-0\rangle \\
&= \langle 4, -3, 1\rangle \\
\ell(t) &= (1, 2, 0) + t(4, -3, 1) \\
&= (1 + 4t, 2 - 3t, t)
\end{aligned}$$

3.1 Dot Product

Definition 3.2: Dot Product

The dot product of two vectors $\vec{v} = \langle v_1, v_2\rangle$ and $\vec{w} = \langle w_1, w_2\rangle$ is defined as:

$$\vec{v} \cdot \vec{w} = v_1 w_1 + v_2 w_2$$

Example 3.3.

$$\begin{aligned}
\vec{v} \cdot \vec{v} &= v_1^2 + v_2^2 = \|\vec{v}\|^2 \\
\vec{i} \cdot \vec{j} &= 1 \cdot 0 + 0 \cdot 1 = 0
\end{aligned}$$

Problem 3.2: Dot Product and Angle

Find pairs of vectors with:

- Dot product > 0
- Dot product $= 0$
- Dot product < 0

Solution 3.2

- $\vec{v} \cdot \vec{v} = \|\vec{v}\|^2 > 0$ for any non-zero vector $\vec{v}$.
- $\vec{i} \cdot \vec{j} = 0$ since they are orthogonal.
- $\langle -1, -1 \rangle \cdot \langle 1, 1 \rangle = -1 - 1 = -2 < 0$.

To conclude, the dot products of any two vectors corresponds to the following situations:

- Dot product $= 0$ if and only if the two vectors are orthogonal (perpendicular).
- Dot product > 0 if the angle between the two vectors is acute (less than 90 degrees).
- Dot product < 0 if the angle between the two vectors is obtuse (greater than 90 degrees).

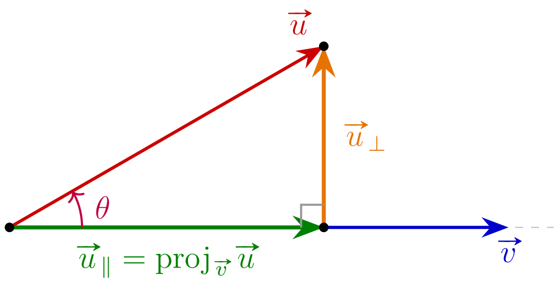
This is because $\vec{v} \cdot \vec{w} = \|\vec{v}\| \|\vec{w}\| \cos\theta$, where θ is the angle between the vectors. Hence,

$$\theta = \arccos\left(\frac{\vec{v} \cdot \vec{w}}{\|\vec{v}\| \|\vec{w}\|}\right)$$

Definition 3.3: Orthogonal Projection

Write $\vec{u} = \vec{u}_\parallel + \vec{u}_\perp$, where $\vec{u}_\parallel \parallel \vec{v}$ and $\vec{u}_\perp \perp \vec{u}_\parallel$. Then, $\vec{u}_\parallel$ is called the **orthogonal projection** of $\vec{u}$ onto $\vec{v}$.

Example 3.4. Below is an illustration of orthogonal projection:



By this, we could see that

$$\vec{u} = \vec{u}_\parallel + \vec{u}_\perp$$

Then, we solve for $\vec{u}_\parallel$ and $\vec{u}_\perp$.

Taking the dot product with $\vec{v}$:

$$\begin{aligned}\vec{u}\cdot\vec{v} &= (\vec{u}_{\parallel}+\vec{u}_{\perp})\cdot\vec{v}\\ &= \vec{u}_{\parallel}\cdot\vec{v}+\vec{u}_{\perp}\cdot\vec{v}\\ &= \vec{u}_{\parallel}\cdot\vec{v}+0 \quad (\text{since } \vec{u}_{\perp}\perp\vec{v})\\ &= \vec{u}_{\parallel}\cdot\vec{v}\end{aligned}$$

Let $\vec{e}_v = \dfrac{1}{\|\vec{v}\|}\vec{v}$ be the unit vector in the direction of $\vec{v}$, and write $\vec{u}_{\parallel} = \lambda\vec{e}_v$. Then:

$$\begin{aligned}\vec{u}\cdot\vec{v} &= (\lambda\vec{e}_v)\cdot(\|\vec{v}\|\,\vec{e}_v)\\ &= \lambda\,\|\vec{v}\|\,(\vec{e}_v\cdot\vec{e}_v)\\ &= \lambda\,\|\vec{v}\|\,\|\vec{e}_v\|^2\\ &= \lambda\,\|\vec{v}\| \quad (\text{since } \|\vec{e}_v\|=1)\end{aligned}$$

Solving for λ:

$$\lambda = \frac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|}$$

Thus, the parallel projection vector $\vec{u}_{\parallel}$ and the orthogonal component $\vec{u}_{\perp}$ are given by:

$$\begin{aligned}\vec{u}_{\parallel} &= \left(\frac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|}\right)\vec{e}_v = \frac{\vec{u}\cdot\vec{v}}{\|\vec{v}\|^2}\vec{v} = \frac{\vec{u}\cdot\vec{v}}{\vec{v}\cdot\vec{v}}\vec{v}\\ \vec{u}_{\perp} &= \vec{u}-\vec{u}_{\parallel} = \vec{u}-\frac{\vec{u}\cdot\vec{v}}{\vec{v}\cdot\vec{v}}\vec{v}\end{aligned}$$

Furthermore, the magnitude of the parallel projection is:

$$\left\|\vec{u}_{\parallel}\right\| = \|\vec{u}\|\,|\cos\theta|$$

Problem 3.3: Vector Projection

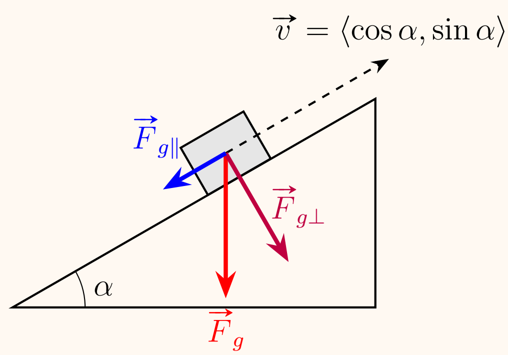
Find the projection of $\vec{u} = \langle 3,2,2\rangle$ onto $\vec{v}\,\langle 1,-2,0\rangle$

Solution 3.3

$$
\begin{aligned}
\vec{u}_{\parallel} &= \frac{3\cdot 1 + 2\cdot(-1) + 2\cdot 0}{1^2 + (-1)^2 + 0^2}\,\langle 1, -1, 0\rangle \\
&= \frac{1}{2}\,\langle 1, -1, 0\rangle \\
&= \left\langle \frac{1}{2}, -\frac{1}{2}, 0 \right\rangle
\end{aligned}
$$

Problem 3.4: Acceleration on an Inclined Plane

Consider a block sliding down an inclined plane with an angle of inclination α.



Solution 3.4

To find the magnitude of acceleration down the incline, we project the gravitational force $\vec{F}_g$ along the incline direction:

$$
\begin{aligned}
\text{acceleration} &= \left\|\vec{F}_{g\parallel}\right\| \\
&= \left\|\vec{F}_g\right\| |\cos\theta| \\
&= \left\|\vec{F}_g\right\| \sin\alpha = 9.8\sin\alpha \quad (\mathrm{m/s^2})
\end{aligned}
$$

Problem 3.5: Vector Projection

Find the projection of $\vec{u} = \langle 3, 2, 2\rangle$ onto $\vec{v} = \langle 1, -1, 0\rangle$.

Solution 3.5

Using the vector projection formula:

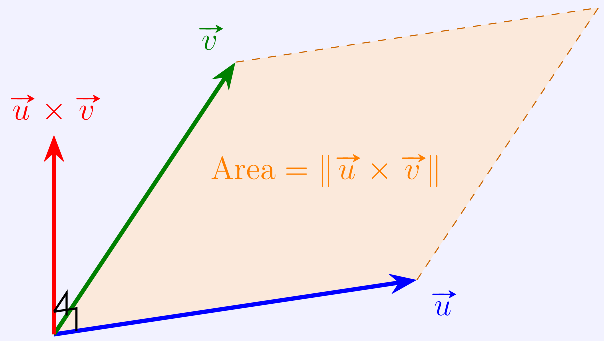
$$\begin{aligned}
\vec{u}_{\parallel} &= \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v} \\
&= \frac{3(1) + 2(-1) + 2(0)}{1^2 + (-1)^2 + 0^2} \langle 1, -1, 0 \rangle \\
&= \frac{3 - 2 + 0}{1 + 1 + 0} \langle 1, -1, 0 \rangle \\
&= \frac{1}{2} \langle 1, -1, 0 \rangle \\
&= \left\langle \frac{1}{2}, -\frac{1}{2}, 0 \right\rangle
\end{aligned}$$

3.2 Cross Product

Definition 3.4: Vector Cross Product

Let $\vec{u}, \vec{v} \in \mathbb{R}^3$. The **cross product** $\vec{u} \times \vec{v}$ is a vector in $\mathbb{R}^3$ satisfying the following three geometric properties:

1. **Orthogonality:** $\vec{u} \times \vec{v} \perp \vec{u}$ and $\vec{u} \times \vec{v} \perp \vec{v}$.
2. **Magnitude:** $\|\vec{u} \times \vec{v}\| = \|\vec{u}\| \, \|\vec{v}\| \sin\theta$, which equals the area of the parallelogram formed by $\vec{u}$ and $\vec{v}$.
3. **Orientation:** The ordered triple $(\vec{u}, \vec{v}, \vec{u} \times \vec{v})$ is right-handed (positively oriented).



From Property (2), we have for the standard basis vectors:

$$\left\| \vec{i} \times \vec{i} \right\| = \left\| \vec{i} \right\| \left\| \vec{i} \right\| \sin 0^\circ = 0 \implies \vec{i} \times \vec{i} = \vec{0}$$

And by the Right-Hand Rule:

$$\vec{i} \times \vec{j} = \vec{k}$$